The Islamia University of Bahawalpur

Faculty of computing

Department of Information Technology

Project Proposal

AI Based Resume Screening System with Web Interface

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Project Proposal

Title:

AI Based Resume Screening System with Web Interface

1. Introduction

Recruitment and hiring is a critical process in every organization, yet manually screening resumes (CVs) is **time-consuming and prone to human bias**. With the increasing number of applicants, HR teams often struggle to shortlist the best candidates efficiently.

Solution:
The proposed project is an **AI Based Resume Screening System** which automatically extracts candidate details from resumes, evaluates their skills, education, and experience, and ranks them according to the requirements of the job role. The system uses **AI techniques including NLP (Natural Language Processing) and ML (Machine Learning)** to ensure accurate evaluation. A **web interface** provides ease of use for both administrators and candidates.

Significance:

- Saves time for HR professionals
- Reduces human error and bias
- Ensures fair evaluation
- Demonstrates AI/NLP/ML application in a real-world scenario

2. Problem Statement

Manual screening of resumes is:

- **Time-consuming**: Reviewing hundreds of CVs manually is inefficient
- **Error-prone**: Human bias can affect candidate selection
- **Non-standardized**: Different recruiters may evaluate candidates differently

The project aims to **automate resume screening** using AI, providing an objective and faster solution while integrating a **user-friendly web platform** for upload and report generation.

3. Objectives

1. Design a **web-based platform** for candidate resume submission and admin management.
2. Use **NLP techniques** to parse resumes and extract information like skills, education, and experience.
3. Implement **AI-based scoring and ranking** for candidates based on job role requirements.

4. Develop an **admin panel** for HR to manage job roles, view candidate rankings, and generate reports.
5. Provide **secure login system** for candidates and admins.
6. Store data in a **database** for persistence and reporting.

4. Scope of the Project

- Applicable for **universities, IT companies, startups, and HR departments**.
- Can be extended to **resume recommendation systems** in the future.
- Enhances **AI understanding**, **web development skills**, and **full-stack project experience**.

5. Methodology

Step 1: Resume Upload & Parsing

- Candidates upload resumes (PDF/DOCX).
- Use **Python libraries** (pdfminer.six, python-docx) to extract text.

Step 2: NLP Based Feature Extraction

- Preprocess text: lowercase, remove punctuation, tokenize.
- Extract features:
  - Skills (Python, Java, SQL, etc.)
  - Education (degree, university)
  - Experience (years, companies)
- Libraries: NLTK, spaCy, TextBlob

Step 3: Candidate Scoring

- Compare candidate features with job requirements.
- Calculate **score / ranking** based on:
  - Skill match
  - Experience match
  - Education match
- Optional ML model for better scoring / classification

Step 4: Database Integration

- Store candidates, resumes, job roles, scores.
- Database: **SQLite or MySQL**

Step 5: Web Interface

- **Frontend:** HTML, CSS, JavaScript, Bootstrap
- **Backend:** Python Flask
- Features:
  - Candidate login / upload
  - Admin dashboard
  - View rankings and reports

Step 6: Admin Panel

- Add / edit / delete job roles
- View candidate rankings
- Generate reports

Step 7: Testing & Deployment

- Test with sample resumes
- Validate NLP parsing and scoring
- Deploy locally or on cloud (optional)

7. Tools & Technologies

Layer	Technology / Tool
Frontend	HTML, CSS, JavaScript, Bootstrap
Backend	Python, Flask
Database	SQLite / MySQL
NLP	NLTK, spaCy, TextBlob
Resume Parsing	pdfminer.six, python-docx
ML	Scikit-learn (classification / ranking)

7. Modules

1. **Login / Authentication Module**
   - Admin and candidate login
2. **Candidate Resume Upload Module**
   - Upload PDF/DOCX resumes
3. **Resume Parsing & NLP Module**
   - Extract features from resumes
4. **Candidate Scoring / AI Module**
   - Calculate score / rank candidates
5. **Admin Panel Module**
   - Manage job roles, view candidates, generate reports
6. **Database Module**
   - Store resumes, scores, candidate info

8. Expected Outcomes

- Fully functional **web-based resume screening system**
- Automatic parsing of resumes
- AI-based ranking of candidates
- Admin dashboard with reports and insights
- Reduced manual effort and bias in candidate selection

9. Project Timeline

Week	Tasks
Week 1-2	Requirement analysis, dataset collection, tools setup
Week 3-4	Design database, login system, web templates
Week 5-6	Resume parsing module (NLP)
Week 7-8	Candidate scoring / ranking (AI/ML)

Week	Tasks
Week 9-10	Admin panel, report generation
Week 11	Integration and testing
Week 12	Documentation, presentation, final report

10. Conclusion

The **AI Based Resume Screening System** will automate the process of evaluating candidate resumes, significantly reducing the time and effort required by HR personnel. By combining **NLP for text parsing** and **AI/ML for candidate scoring**, the system ensures **objective, fair, and efficient recruitment**. The integration of a **web interface and database** makes the system user-friendly, scalable, and ready for real-world implementation.

This project provides a **strong AI application example**, enhances skills in **NLP, ML, web development, and database management**, and demonstrates a **full-stack approach** suitable for academic evaluation and future expansion.